Analyzing Dataset Uncertainty and Calibration Behavior in Classical Medical Binary Classifiers

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***Abstract*—This study analyzes uncertainty and calibration behavior in classical binary classification models using the Pima Indians Diabetes dataset and the Breast Cancer Wisconsin Diagnostic dataset. Logistic Regression, Support Vector Machine, Random Forest, and Multilayer Perceptron models are evaluated across different feature subsets, hyperparameter configurations, and post-hoc calibration methods. Dataset-level uncertainty is decomposed into aleatoric and epistemic components using bootstrap ensembles, while model reliability is evaluated using Expected Calibration Error, Negative Log-Likelihood, Brier Score, predictive entropy, and AUC-ROC. The results show that the Pima Indians Diabetes dataset has higher aleatoric uncertainty and lower predictive performance, suggesting that intrinsic noise and class ambiguity limit model learning. In contrast, the Breast Cancer Wisconsin dataset has higher feature redundancy but achieves better performance and lower uncertainty, indicating that redundant features can still be useful when they preserve discriminative information. Feature subsets selected using Mutual Information and mRMR provide the best trade-off between performance and uncertainty. Calibration improves ECE in many cases, especially for MLP and Logistic Regression, but does not always improve NLL. Overall, the results indicate that dataset uncertainty, feature structure, model architecture, and calibration method should be considered together when developing reliable medical classifiers.**

***Index Terms*—Uncertainty quantification, model calibration, binary classification, medical datasets, aleatoric uncertainty, epistemic uncertainty, bootstrap ensembles, feature selection.**

I. Introduction

In situations where model interpretability is important, classical models such as Logistic Regression, Support Vector Machines (SVM), Random Forests, and Multilayer Perceptrons (MLP) remain highly relevant because they allow an easier understanding of the reasons behind a model's decision. Model interpretability is especially important in fields such as medicine, where incorrect decisions can have serious consequences. In such situations, it is not only important for the model to make the correct decision; it is also essential that physicians, researchers, and patients can understand why the model made that decision, and how certain or uncertain the model is about it.

For many years, accuracy was used as the main metric for evaluating model quality. However, as the only measure of performance, accuracy is fundamentally incomplete. To illustrate this, we can return to the case in which a model assists in establishing a diagnosis and consider two scenarios. In the first case, the model's output is:

The person is not sick.

In the second case, the model's output is:

The person is not sick. Certainty: 51%.

This small difference in the responses makes a drastic difference, especially when the model operates in an environment where decision-making carries great responsibility. The first model classifies the person as healthy without providing any indication of how confident it is in its decision, while the second model provides an additional piece of information of immeasurable importance. In high-risk applications, with medicine being a very clear example, model certainty is not a redundant detail but an essential piece of information. This and many similar situations are the motivation for studying *Uncertainty Quantification* (UQ) in machine learning models.

Uncertainty in machine learning models originates from two fundamentally different sources. The first source is the data with which the model works, which causes *aleatoric uncertainty*. This type of uncertainty is irreducible because it comes from inherent variability, noise, or ambiguity in the data itself. The second source is uncertainty that arises from a lack of knowledge or insufficient data. This type of uncertainty is reducible by nature and is called *epistemic uncertainty*.

In order to measure and improve uncertainty, this research applies post-hoc calibration methods. These are methods applied after model training with the aim of improving the reliability of probabilistic predictions.

This research aims to answer five key questions:

1) How does the nature of the dataset differ in terms of the dominant type of uncertainty?
2) How does each type of dataset affect the learning capabilities of classical models?
3) How can feature engineering be performed on a given dataset in order to minimize uncertainty?
4) How much does model selection matter in the context of uncertainty and calibration?
5) How does the application of calibration methods affect different models and datasets?

This paper contributes to a better understanding of how decisions made during model development, including feature

selection, hyperparameter tuning, architecture selection, and the application of calibration methods, affect uncertainty in classical binary classifiers.

II. Background

A. Uncertainty in Machine Learning

Uncertainty in machine learning models is a phenomenon that can originate from different sources. Studying these sources is of crucial importance for understanding how a model makes decisions and, based on the results, determining how the model can be improved. There are two main types of uncertainty: *aleatoric uncertainty* and *epistemic uncertainty*.

Aleatoric uncertainty, also known as statistical uncertainty, originates from the structure of the data itself and refers to the randomness present in the data-generating process. Its cause is not related to weaknesses of the model or to an insufficient amount of data. Instead, it represents an inherent part of the nature of the problem.

One example closely related to the data used in this research is the measurement of glucose levels or blood pressure. The same values can be measured differently at different points in time, under different physiological conditions, or due to variability between patients. This type of uncertainty is *irreducible*, meaning that it cannot be eliminated by using a better model or by adding more data.

Epistemic uncertainty, on the other hand, is the result of a lack of knowledge or an insufficient amount of data available to the model. This type of uncertainty is *reducible*, meaning that collecting more relevant data, selecting better features, or using a more sophisticated model can help reduce it.

The decomposition of total uncertainty into aleatoric and epistemic components is performed using the *bootstrap ensemble* method. A total of M models are trained, where each model receives a different training subset generated through bootstrapping. For each test example x, each of the M models generates its own probabilistic prediction:

$$p_m(y = 1 \mid x) \tag{1}$$

After this, the total uncertainty is measured using the entropy of the mean ensemble prediction:

$$U_{\text{total}}(x) = H\left(\bar{p}(y = 1 \mid x)\right) \tag{2}$$

where $\bar{p}$ represents the mean probability predicted by the M models:

$$\bar{p}(y = 1 \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_m(y = 1 \mid x) \tag{3}$$

Aleatoric uncertainty is obtained as the average entropy of the individual model predictions:

$$U_{\text{alea}}(x) = \frac{1}{M} \sum_{m=1}^{M} H\left(p_m(y = 1 \mid x)\right) \tag{4}$$

The intuition behind this formulation is the following: if every model remains uncertain for the same example, regardless of the training subset on which it was trained, then the uncertainty originates from the data itself, namely from the noise or ambiguity inherent in that example. This represents the aleatoric component.

Epistemic uncertainty is then obtained as the difference between total and aleatoric uncertainty:

$$U_{\text{epis}}(x) = U_{\text{total}}(x) - U_{\text{alea}}(x) \tag{5}$$

This component represents the disagreement among the models in the ensemble. If the models produce noticeably different predictions for the same example, this indicates that the model is uncertain due to insufficient knowledge or insufficient data, which corresponds to the epistemic component.

Therefore, the formula used to compute total model uncertainty is a consequence of the bootstrap-based uncertainty decomposition.

B. Datasets

In this research, two medical datasets are used, representing opposite extremes on the spectrum of aleatoric and epistemic uncertainty.

The Pima Indians Diabetes dataset contains clinical measurements such as glucose level, blood pressure, and body mass index. These quantities are naturally variable and susceptible to noise, which makes this dataset predominantly aleatoric [2].

On the other hand, the Breast Cancer Winconsin dataset contains measurements obtained from medical images of tumor tissue using precise and calibrated technology. Therefore, these measurements are inherently more stable, and the uncertainty of the model depends primarily on epistemic factors [3].

C. Model Calibration and Conformal Prediction

Model calibration refers to the process of adjusting the predicted probabilities of a model so that they reflect the true likelihood of an event. A visual tool used to evaluate model calibration is the *reliability diagram*, which compares the mean predicted probability with the empirical accuracy within predefined probability intervals. A perfectly calibrated model produces a reliability diagram that lies on the diagonal line, while deviations from the diagonal indicate systematic calibration error.

Post-hoc calibration methods are applied after the model has already been trained. For these methods, a separate dataset is reserved, known as the *calibration set*. This calibration set is used to learn a transformation that maps the original, uncalibrated model outputs into better-calibrated probability estimates [1].

Platt Scaling is essentially a logistic regression model applied on top of a classifier that produces raw scores or logits. It is represented as:

$$p_{\text{cal}} = \frac{1}{1 + \exp(-(a \cdot s + b))} \tag{6}$$

where s represents the uncalibrated model score, while a and b are parameters estimated by maximizing the log-likelihood on the calibration set. This is equivalent to fitting logistic regression on the uncalibrated outputs of the model [1].

Isotonic calibration, also known as isotonic regression, is a non-parametric post-processing technique used to adjust the predicted probabilities of a machine learning classifier so that they better reflect true empirical likelihoods. Unlike Platt Scaling, isotonic calibration does not assume a specific parametric form of the calibration function, which makes it more flexible [4].

Histogram binning is a calibration technique that organizes continuous predicted probabilities into discrete intervals, called bins. It is used to analyze probability distribution patterns and reduce the complexity of calibration. For an uncalibrated prediction that falls into bin b, the calibrated probability is computed as [5] :

$$p_{\text{cal}} = \frac{\text{number of positive examples in bin } b}{\text{total number of examples in bin } b} \tag{7}$$

D. Uncertainty and Calibration Metrics

In this project, five metrics were used to evaluate uncertainty and calibration: predictive entropy, Expected Calibration Error (ECE), Brier Score, Negative Log-Likelihood (NLL), and AUC-ROC.

Predictive entropy measures the uncertainty of a probabilistic prediction. For binary classification, if the model predicts probability p for the positive class, predictive entropy is computed as:

$$H(p) = -p \cdot \log(p) - (1-p) \cdot \log(1-p) \tag{8}$$

Predictive entropy reaches its maximum value when $p = 0.5$, which means that the model is completely uncertain between the two classes. For the entire test dataset, the mean predictive entropy is computed as:

$$\bar{H} = \frac{1}{n} \sum_{i=1}^{n} H(p_i) \tag{9}$$

where n is the total number of test examples and p_i is the predicted probability for the positive class for the i-th example.

Expected Calibration Error (ECE) measures the difference between the model's predicted confidence and its empirical accuracy. It is computed as:

$$ECE = \sum_{b=1}^{B} \frac{n_b}{n} \cdot |acc(b) - conf(b)| \tag{10}$$

where $acc(b)$ is the empirical accuracy in bin b, $conf(b)$ is the mean predicted confidence in bin b, n_b is the number of examples in bin b, n is the total number of examples, and B is the total number of bins. An ECE value of zero indicates perfect calibration. Values greater than zero indicate calibration error, where larger values correspond to stronger miscalibration.

The *Brier Score* measures the mean squared difference between the predicted probabilities and the true labels. It is computed as:

$$BS = \frac{1}{n} \sum_{i=1}^{n} (\hat{p}_i - y_i)^2 \tag{11}$$

where $\hat{p}_i$ is the predicted probability for the i-th example and y_i is the true class label. A lower Brier Score indicates that the predictions are both more accurate and more reliable.

Negative Log-Likelihood (NLL) evaluates the quality of probabilistic predictions and is especially sensitive to extremely confident but incorrect predictions. For example, a model that assigns a probability close to zero to an example that actually belongs to the positive class will be heavily penalized. NLL is computed as:

$$NLL = -\frac{1}{n} \sum_{i=1}^{n} [y_i \cdot \log(\hat{p}_i) + (1-y_i) \cdot \log(1-\hat{p}_i)] \tag{12}$$

Finally, *AUC-ROC* measures how well the model distinguishes positive examples from negative examples, regardless of whether the predicted probabilities are well calibrated. It is computed as the area under the Receiver Operating Characteristic (ROC) curve, which represents the relationship between sensitivity and specificity across different decision thresholds. A value of 1.0 indicates perfect discrimination, while a value of 0.5 indicates random guessing.

III. Related Works

A. Reliable probability estimation and calibration

Results of previously conducted research have confirmed that high model accuracy does not necessarily mean that its probabilistic predictions are well calibrated. This is precisely the reason why, in addition to classification accuracy, it is important to analyze how confident a model is in its predictions. Niculescu-Mizil and Caruana studied the quality of probabilistic predictions of various supervised learning algorithms, including SVM, neural networks, decision trees, Random Forest, Naive Bayes, and Logistic Regression. Their results showed that different models produce different forms of probability distortion, and that methods such as Platt Scaling and Isotonic Regression are effective ways to improve calibration [7]. The connection between our research and the aforementioned work lies in the fact that the objective of our research is also the calibration and reliability of probabilistic outputs. However, our research additionally analyzes various uncertainty measures with a focus on binary classification.

B. Calibration of modern neural networks

Research on the calibration of modern neural networks has shown that high predictive accuracy does not guarantee reliable confidence estimates. The study confirmed that modern neural networks can achieve high accuracy while still being poorly calibrated, which supports the main motivation of our research [8]. This research is relevant to the problem studied in our

project because it emphasizes that accuracy alone is not sufficient for evaluating the quality of a classifier. While the aforementioned study primarily focuses on modern neural network architectures, our focus is placed on classical machine learning models and on comparing their behavior in terms of uncertainty and calibration in binary classification tasks.

C. Ensemble-based uncertainty estimation

In this study, the approach is based on training multiple models and combining their predictions in order to obtain more reliable uncertainty estimates [9]. In our case, bootstrap ensembles are used to decompose total uncertainty into aleatoric and epistemic components. The main difference is that our research applies uncertainty decomposition to classical binary classifiers, while the aforementioned study focuses on deep neural networks.

IV. Methodology

This study investigates how dataset characteristics, feature engineering, model selection, hyperparameter configuration, and post-hoc calibration affect uncertainty in classical binary classification models. Therefore, the experimental pipeline consists of six main stages: dataset preprocessing, feature subset generation, model training, calibration, and uncertainty decomposition.

A. Datasets and Preprocessing

Two medical binary classification datasets are used in this study: the Pima Indians Diabetes dataset and the Breast Cancer Winconsin Diagnostic dataset. These datasets were selected because they have different feature structures and are expected to exhibit different uncertainty characteristics.

The Pima Indians Diabetes dataset contains 768 samples, 8 clinical features, and a binary target. The features include Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age. Since several clinical variables contain biologically invalid zero values, the values of zero in Glucose, BloodPressure, SkinThickness, Insulin, and BMI are treated as missing values. These values are replaced with `NaN` and then median-imputed. The imputation statistics are computed only on the training split and are then applied to the validation, calibration, and test splits. This is to prevent information leakage.

The Breast Cancer Winconsin Diagnostic dataset contains 569 samples, 30 numerical features, and a binary target. The features are image-derived nuclear measurements, including mean, standard error, and worst values of radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. Since this dataset contains no missing values, no imputation is applied.

For both datasets, a four-way stratified split is used: 60% training, 15% validation, 15% calibration, 10% test. The current experiment uses a fixed random seed to ensure reproducibility.

B. Feature Subset Generation

Instead of using only the full feature set, several systematic feature subset families are generated. These subsets allow us to compare whether uncertainty is reduced by removing noisy, redundant, or less informative features.

Let the complete feature set be denoted as:

$$F_{\text{all}} = \{f_1, f_2, \ldots, f_d\}, \tag{13}$$

where d is the number of features in the dataset. This dataset is then transformed into 5 feature subset families:

1) Family A: All Features: The first subset contains all available features. This serves as the baseline representation and is used to determine whether feature reduction improves or worsens performance, calibration, and uncertainty.

2) Family B: Top-k Statistical Feature Subsets: The second family contains top-k feature subsets selected using statistical methods. Two ranking methods are used: Mutual Information (MI) and Minimum Redundancy Maximum Relevance (mRMR).

Mutual Information ranks features according to their dependence with the target variable. Features with higher MI scores are considered more informative for classification.

$$MI(F_j, Y) = \sum_{f_j} \sum_{y} p(f_j, y) \log \left(\frac{p(f_j, y)}{p(f_j)p(y)} \right). \tag{14}$$

The mRMR method is used to select features that are both relevant to the target and minimally redundant with each other. At each step, the selected feature is the one that maximizes:

$$\text{score}(f) = MI(f, y) - \frac{1}{|S|} \sum_{f_j \in S} MI(f, f_j), \tag{15}$$

where S is the set of already selected features, $MI(f, y)$ measures relevance to the target, and $MI(f, f_j)$ measures redundancy with previously selected features.

The values of k are selected from:

$$k \in \{3, 5, 10, 15\}, \tag{16}$$

with the value capped by the total number of features in the dataset.

3) Family C: Correlation-Filtered Feature Subsets: The third family removes highly correlated features. This is especially important for the Breast Cancer Winconsin dataset, where many features are derived from related morphological measurements. The absolute Pearson correlation matrix is computed on the training set. If two features have an absolute correlation greater than a threshold, the feature with lower Mutual Information with the target is removed.

The thresholds used are:

$$\rho \in \{0.80, 0.90, 0.95\}. \tag{17}$$

This produces feature subsets that reduce redundancy while preserving informative variables.

4) Family D: Domain-Based Feature Groups: This family consists of feature groups defined using domain knowledge. For the Pima Indians Diabetes dataset, the domain groups include metabolic features, cardiovascular features, demographic/pregnancy features, skin/body composition features, genetic risk proxy features, and a metabolic-plus-risk group.

For the Breast Cancer Winconsin dataset, domain groups are based on morphological characteristics such as size and shape, texture and symmetry, concavity, worst-size features, worst-shape features, error features, and a core discriminative group.

These subsets are included to make the uncertainty analysis more interpretable. They allow us to examine whether uncertainty is associated with specific clinical or morphological sources of information.

5) Family E: Random Feature Subsets: Random feature subsets are generated as a control group. These subsets are used to verify whether statistically selected or domain-based subsets perform better than chance. However, random subsets are excluded from the coarse experimental search to reduce computational cost.

C. Model Training

Four classical binary classification models are evaluated:

1) Logistic Regression,
2) Support Vector Machine,
3) Random Forest,
4) Multilayer Perceptron.

Each model is trained across multiple feature subsets and hyperparameter configurations. For every experiment, the model is trained on the training split, evaluated on the validation and test splits, and then calibrated using the calibration split. The choice of hyperparameters are given in the Appendix.

D. Hyperparameter Selection

Each model is evaluated over a coarse hyperparameter grid. The goal of hyperparameter selection is not only to maximize predictive performance, but also to select models with reliable probabilistic behavior. Therefore, validation performance is summarized using a composite score:

$$S_c = AUC - 0.5 \cdot ECE - 0.1 \cdot NLL. \tag{18}$$

This score rewards discrimination through AUC while penalizing poor calibration through Expected Calibration Error and Negative Log-Likelihood. A model that achieves high AUC but produces unreliable probability estimates can therefore receive a lower composite score.

E. Evaluation Metrics

Each model is evaluated using both classification and probabilistic metrics. The classification metrics include accuracy, precision, recall, F1-score, and AUC-ROC. The uncertainty and calibration metrics include Brier Score, Negative Log-Likelihood, Expected Calibration Error, Maximum Calibration Error, and predictive entropy.

F. Post-Hoc Calibration

Post-hoc calibration is applied after the base model has already been trained. The base classifier is not re-trained during calibration. Instead, calibration methods learn a transformation from the original predicted probabilities to calibrated probabilities using only the calibration set. The calibrated predictions are then evaluated on the test set. The calibration methods include platt scaling, isotonic calibration, and histogram binning. The hyperparameters chosen for each calibration methods are described in the Appendix.

G. Bootstrap-Based Uncertainty Decomposition

Calibration metrics indicate whether predicted probabilities are reliable, but they do not directly separate aleatoric and epistemic uncertainty. Therefore, a bootstrap ensemble is used to approximate the decomposition of total uncertainty.

For each selected model architecture and feature subset, the best hyperparameter configuration is selected according to the validation composite score. Then, $M = 30$ bootstrap models are trained. Each bootstrap model is trained on a sample drawn with replacement from the original training set. The ensemble mean prediction is computed as:

$$\bar{p}(y = 1 \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_m(y = 1 \mid x). \tag{19}$$

Total predictive uncertainty is measured as the entropy of the ensemble mean:

$$U_{\text{total}}(x) = H\left(\bar{p}(y = 1 \mid x)\right). \tag{20}$$

Aleatoric uncertainty is approximated as the average entropy of the individual bootstrap predictions:

$$U_{\text{alea}}(x) = \frac{1}{M} \sum_{m=1}^{M} H\left(p_m(y = 1 \mid x)\right). \tag{21}$$

This quantity is high when all bootstrap models are individually uncertain about the same sample. Such uncertainty is interpreted as uncertainty caused by ambiguity or noise in the data.

Epistemic uncertainty is computed as the difference between total uncertainty and aleatoric uncertainty:

$$U_{\text{epis}}(x) = \max\left(U_{\text{total}}(x) - U_{\text{alea}}(x), 0\right). \tag{22}$$

This quantity measures disagreement among bootstrap models. If different bootstrap models produce different confident predictions, epistemic uncertainty becomes larger.

The variance of predicted probabilities is also computed:

$$VAR_p(x) = \frac{1}{M} \sum_{m=1}^{M} \left(p_m(y = 1 \mid x) - \bar{p}(y = 1 \mid x)\right)^2. \tag{23}$$

Dataset-level uncertainty summaries are then obtained by averaging sample-level values over n resulting bootstrapped models on the test set:

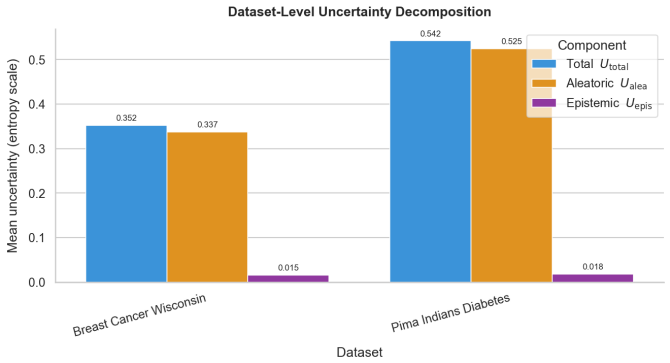


Fig. 1. Dataset-level uncertainty decomposition measured by bootstrapping the best model of each model type and feature subset pair.

$$\bar{U}_{\text{total}} = \frac{1}{n}\sum_{i=1}^{n} U_{\text{total}}(x_i), \tag{24}$$

$$\bar{U}_{\text{alea}} = \frac{1}{n}\sum_{i=1}^{n} U_{\text{alea}}(x_i), \tag{25}$$

$$\bar{U}_{\text{epis}} = \frac{1}{n}\sum_{i=1}^{n} U_{\text{epis}}(x_i). \tag{26}$$

The relative aleatoric and epistemic ratios are computed as:

$$R_{\text{alea}} = \frac{\bar{U}_{\text{alea}}}{\bar{U}_{\text{total}} + \epsilon}, \tag{27}$$

$$R_{\text{epis}} = \frac{\bar{U}_{\text{epis}}}{\bar{U}_{\text{total}} + \epsilon}, \tag{28}$$

where ϵ is a small constant used for numerical stability.

V. Results

A. Dataset Uncertainty Analysis

Using bootstrapping to decompose dataset-level uncertainty as described in IV-G, the aleatoric and epistemic uncertainty levels of the two datasets were measured. As can be seen in Figure 1, the Pima Indians dataset yields approximately 0.2 higher aleatoric uncertainty score than the corresponding Breast Cancer Winconsin dataset. This empirical result directly supports the claim that the measurement tools used to acquire the data and the noisy nature of the data itself corresponds to the overall uncertainty of the underlying dataset.

As shown in Figure 2, the Pima Indians dataset shows significantly less inter-correlation between features than Breast Cancer Winconsin dataset. Empirically, this means that the Breast Cancer Winconsin dataset has significantly higher feature redundancy than the Pima Indians dataset.

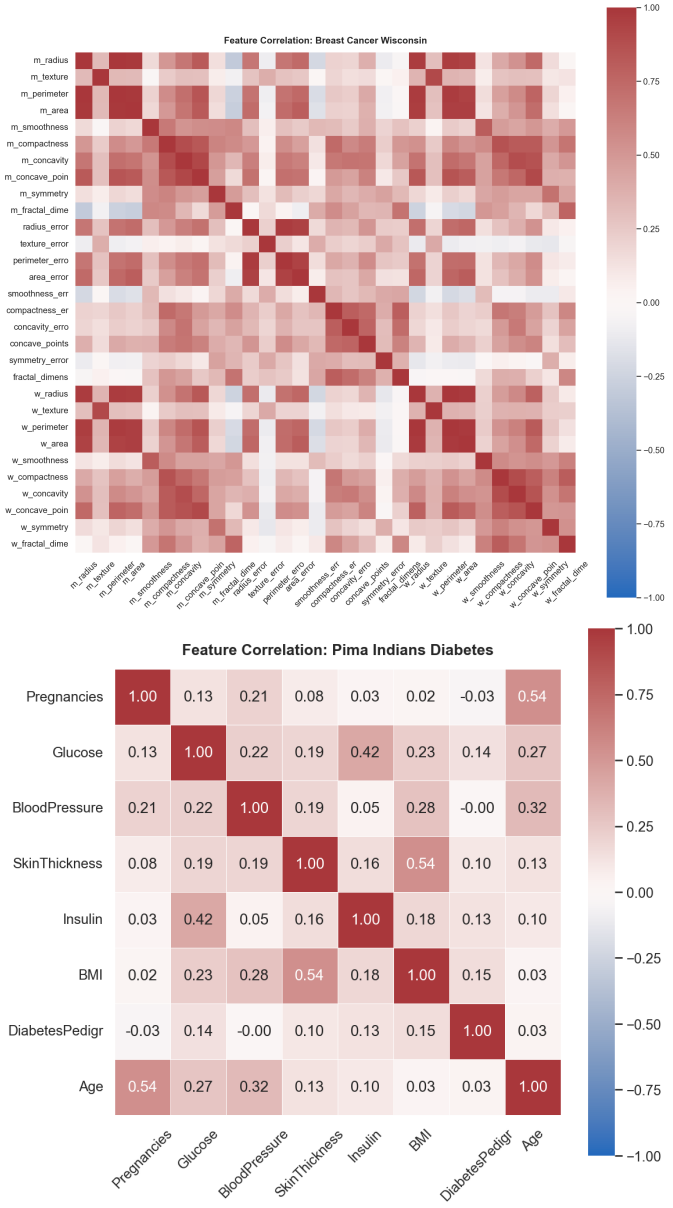


Fig. 2. Correlation map between internal features in each dataset.

B. Learning Performance

In order to measure the correlation between learning performance with its underlying dataset, two distinct figures were plotted. The first figure, Figure 3, depicts performance distribution by dataset and model, while the second figure, Figure 4, depicts model performance vs aleatoric uncertainty color-coded by dataset. All scores are quantified with the uncalibrated models and the test subset to measure performance in real deployment.

Although the Breast Cancer Winconsin dataset empirically has higher levels of feature redundancy, the results shown in Figure 3 seems to indicate that classical model were able to draw a clearer decision boundary in comparison to the Pima Indian dataset that has been analyzed to have higher

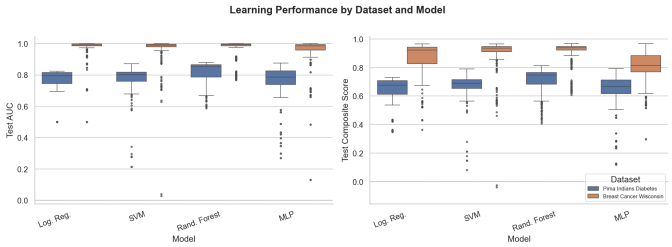


Fig. 3. Learning Performance across all experiments separated by uncalibrated model and dataset type.

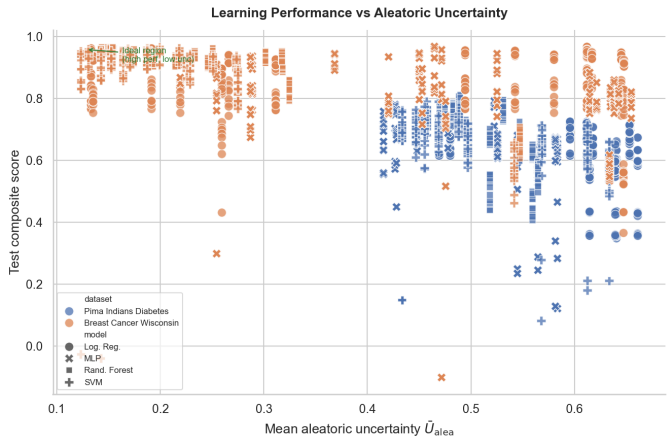


Fig. 4. Uncertainty performance across all experiments separated by uncalibrated model and dataset type.

aleatoric uncertainty score. A reason for this might be that even though features are inter-correlated, the amount of information gathered from its large numbers of features is enough for the model to not overfit.

Depicted by Figure 4, the Pima Indians dataset correspond to lower composite score as well as higher aleatoric uncertainty score across all model-feature pair in comparison to its counterpart. This is well expected given the results in V-A. Despite this, it is worth noting that many of the model-feature pairs of the Breast Cancer Winconsin dataset have high aleatoric uncertainty score, this will be explained in V-C

C. Feature Selection

In part to understand how feature selection affects the model's performance and uncertainty, graphs depicting the relation between the number of features and performance scores are plotted in Figures 5, 6, and 7. The figures are measured by splitting the models into feature family types and averaging its performance scores.

Figures 5 and 6 both show that the amount of information does not automatically infer higher performance, but instead the statistical properties of individual features do. Reinforcing this observation, figure 7 shows that feature family B, choosing top-k features based on Mutual Information and mRMR, simultaneously shows the highest mean performance and lowest mean aleatoric uncertainty scores on both dataset.

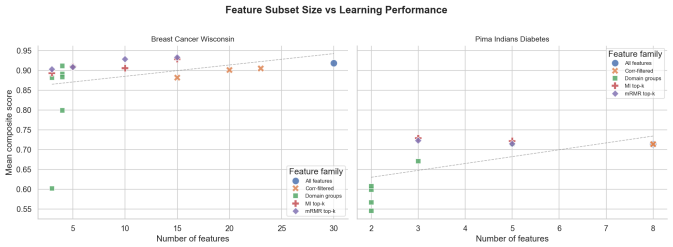


Fig. 5. Feature family vs Mean composite score.

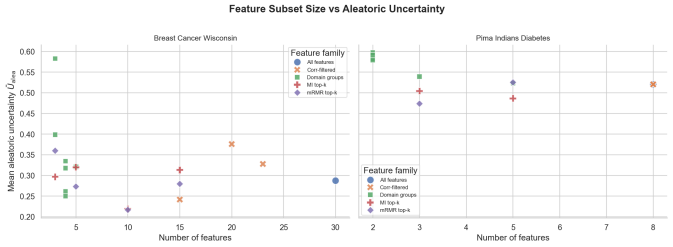


Fig. 6. Feature family vs Mean uncertainty score.

A side note here is that some feature selection family for the Breast Cancer Winconsin also result in high aleatoric uncertainty score. This fact is reflected back to Figure 4, where some resulting points for the dataset has above 0.4 aleatoric uncertainty score.

Observation of the results show that feature selection based on statistical measurement result in overall better performance across multiple model architecture as opposed to correlation-filtered, data domain, or random based selections.

D. Model Selection

Model selection is arguably the most important part in solving machine learning. The goal of this experiment is to simultaneously analyze which model performs the best and least uncertain about its prediction. Here, Figure 8 depicts the correlation between model architecture and its uncertainty score. Furthermore, Figure 9 measures the stability of the model itself. All figures are plotted with the model architecture across all feature subsets and hyperparameter configurations.

Figure 8 shows that the answer to which model performs the most certain depends on the nature of the data itself. SVM is best in predicting the Breast Cancer Winconsin dataset with certainty, by having the least overall score, lowest mean, and smallest variance. On the other hand, while not having the lowest mean or variance, MLP have the lowest overall uncertainty score for the Pima Indians dataset.

Interestingly, Figure 9 show that Random Forest have the most stable composite scorea across all configuration of features and hyperparameter tested.

Although it is clear that different data requires different approaches, further analysis of the geometrical and statistical properties of the underlying data have to be made to conclude the reason of the performance of these models.

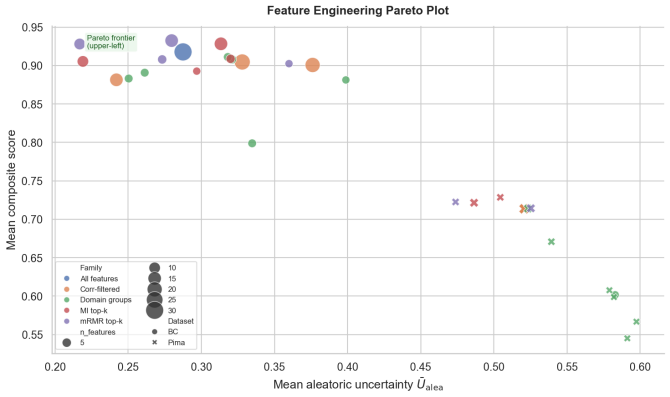


Fig. 7. Pareto plot of mean composite vs mean uncertainty score across all feature family.

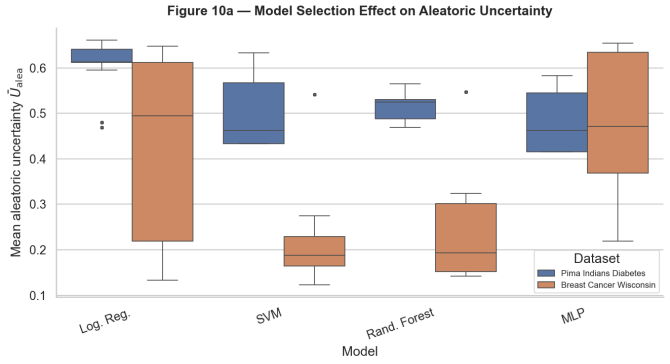


Fig. 8. Model architecture vs Mean uncertainty score.

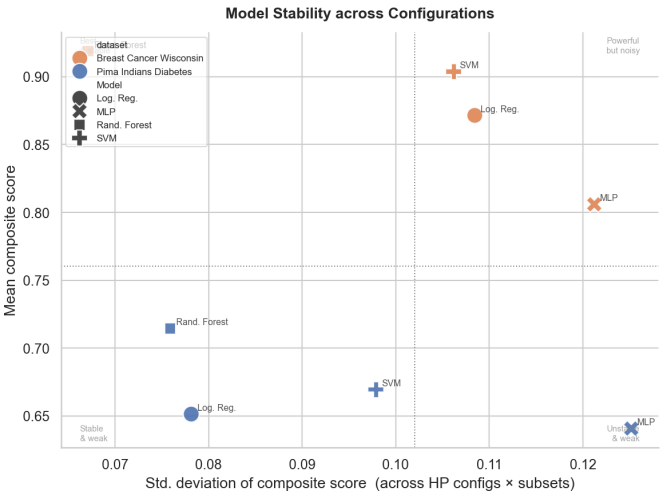


Fig. 9. Composite score stability across all model architecture.

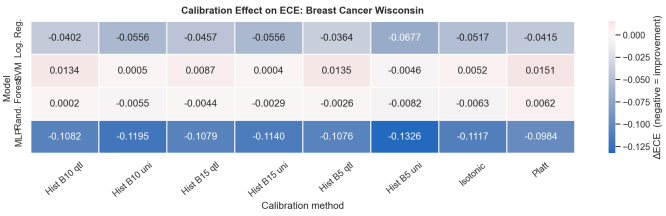


Fig. 10. Calibration effect on ECE for the Breast Cancer Wisconsin dataset. Negative values indicate improvement.

E. Calibration Method Selection

To analyze calibration effectiveness, the change in Expected Calibration Error was computed for each model-calibration pair:

$$\Delta ECE = ECE_{\text{calibrated}} - ECE_{\text{original}}. \tag{29}$$

A negative ΔECE indicates that calibration improves model reliability, while a positive value indicates that calibration worsens it.

As shown in Figure 10, calibration generally improves ECE for the Breast Cancer Wisconsin dataset. The strongest improvement appears in MLP, where all calibration methods reduce ECE. Histogram binning with 5 bins using uniform binning gives the largest improvement, with $\Delta ECE = -0.1326$. Logistic Regression also benefits consistently, while Random Forest shows only small changes.

Figure 11 shows that calibration has a less consistent effect on the Pima Indians Diabetes dataset. Logistic Regression and MLP mostly improve after calibration, but SVM and Random Forest show mixed results. This indicates that calibration does not always improve reliability and depends on the model-dataset pair.

Figure 12 shows that many calibrated models improve ECE while worsening NLL. This means that calibration can make predicted probabilities more reliable according to ECE, but not necessarily better according to likelihood. This is expected because ECE measures confidence reliability, while NLL strongly penalizes confident incorrect predictions.

As shown in Figure 13, the Pima Indians Diabetes dataset generally has more negative median ΔECE values, meaning that calibration often improves its ECE score. However, it also has more outliers, indicating less stable calibration behavior. In comparison, the Breast Cancer Wisconsin dataset has median values closer to zero, suggesting that many models were already relatively well calibrated before post-hoc calibration.

From these results, calibration is useful but not universally effective. Its effect depends on the dataset, model architecture, and calibration method. In general, MLP and Logistic Regression benefit the most from calibration, while Random Forest and SVM show more mixed behavior.

VI. Discussion

From the results, we can observe that dataset characteristics strongly affect model performance, uncertainty, and calibration behavior. Although the Breast Cancer Wisconsin dataset has higher feature redundancy, it still achieves better overall performance than the Pima Indians Diabetes dataset. This suggests that feature redundancy alone does not necessarily reduce model performance. If the correlated features still contain useful class information, the model can still learn a clear decision boundary.

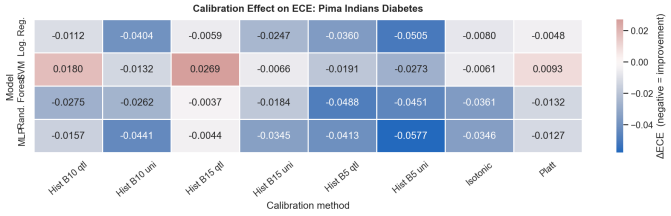


Fig. 11. Calibration effect on ECE for the Pima Indians Diabetes dataset. Negative values indicate improvement.

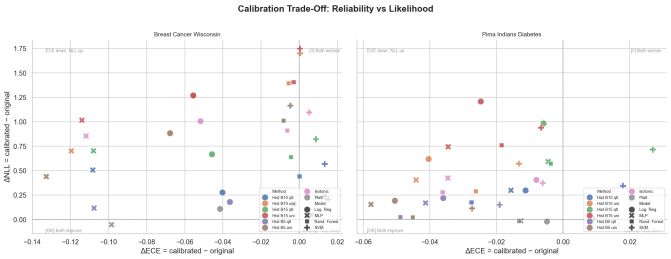


Fig. 12. Calibration trade-off between reliability and likelihood. The x-axis shows ΔECE, while the y-axis shows ΔNLL.

On the other hand, the Pima Indians dataset shows higher aleatoric uncertainty and lower performance across most model configurations. This indicates that the main difficulty of this dataset comes from the noisy and ambiguous nature of the data itself. Since aleatoric uncertainty is related to irreducible uncertainty, better model selection or calibration can improve the result only to a certain extent.

The feature selection results also support this observation. Feature subsets selected using Mutual Information and mRMR generally produce better performance and lower uncertainty compared to other feature families. This means that statistically informative features are more useful than simply using all features or removing features only based on correlation. In this experiment, reducing the number of features does not automatically improve performance. Instead, the quality and relevance of the selected features matter more.

The model selection results show that the best model depends on the dataset. SVM gives the lowest uncertainty for the Breast Cancer Wisconsin dataset, while MLP performs better in terms of uncertainty for the Pima Indians dataset. Random Forest shows more stable behavior across different configurations, even though it does not always give the lowest uncertainty. Therefore, model selection should consider both the best performance score and the stability of the model across different feature and hyperparameter settings.

The calibration results show that post-hoc calibration is useful, but not always effective. MLP and Logistic Regression benefit the most from calibration, while SVM and Random Forest show more mixed results. Furthermore, the comparison between ΔECE and ΔNLL shows that improving ECE does not always improve NLL. This means that calibration can improve probability reliability while worsening likelihood-based performance.

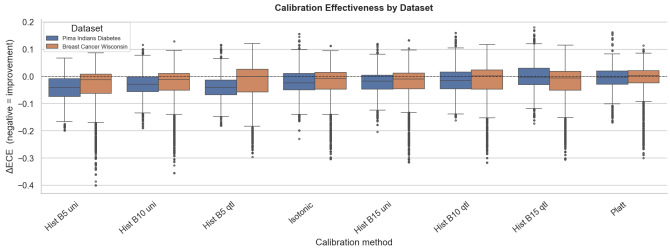


Fig. 13. Calibration effectiveness by dataset. Negative ΔECE values indicate improvement.

Overall, these results suggest that dataset-level uncertainty has a stronger effect on model performance than feature redundancy alone. The Breast Cancer Wisconsin dataset contains more redundant features, but its lower aleatoric uncertainty makes the classification problem easier. In contrast, the Pima Indians dataset has less feature redundancy, but its higher aleatoric uncertainty makes the learning task more difficult.

A. *Limitations*

This study has several limitations. First, only two binary medical datasets were used. Therefore, the results cannot be generalized to all medical classification problems. Second, the uncertainty decomposition is based on bootstrap estimation, so the aleatoric and epistemic scores should be interpreted as empirical approximations rather than exact values.

Another limitation is the size of the calibration split. Since the datasets are relatively small, the calibration set may not always be large enough to learn a stable calibration mapping. This may explain why some calibration methods worsen ECE or NLL in certain model-dataset pairs. Finally, this study only focuses on classical machine learning models and does not compare them with deep learning or Bayesian models.

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APPENDIX A
EXPERIMENTAL HYPERPARAMETERS

This appendix describes the hyperparameter configurations used in the experiments. For each model architecture, a coarse hyperparameter grid was defined. Each model was trained over all feature subsets and all hyperparameter configurations. The best configuration was selected using the validation composite score described in Section IV-G.

A. Model Hyperparameters

TABLE I
HYPERPARAMETER GRID USED FOR LOGISTIC REGRESSION.

Hyperparameter	Values
Penalty	l_2
Regularization strength C	$\{0.01, 0.1, 1, 10\}$
Solver	liblinear
Maximum iterations	1000

For Logistic Regression, the main hyperparameter is the regularization strength C. Smaller values of C apply stronger regularization, while larger values allow the model to fit the training data more freely. The l_2 penalty was used to reduce overfitting and improve numerical stability.

TABLE II
HYPERPARAMETER GRID USED FOR SUPPORT VECTOR MACHINE.

Hyperparameter	Values
Kernel	{linear, rbf}
Regularization strength C	$\{0.1, 1, 10\}$
Kernel coefficient γ	{scale, auto}
Probability output	True

For SVM, both linear and radial basis function kernels were tested. The parameter C controls the trade-off between margin maximization and classification error, while γ controls the influence range of individual samples in the RBF kernel. Probability estimation was enabled so that calibration and uncertainty metrics could be computed.

TABLE III
HYPERPARAMETER GRID USED FOR RANDOM FOREST.

Hyperparameter	Values
Number of trees	$\{100, 200\}$
Maximum depth	$\{None, 5, 10\}$
Minimum samples split	$\{2, 5\}$
Minimum samples leaf	$\{1, 2\}$

For Random Forest, the number of trees controls ensemble size, while maximum depth and minimum sample constraints control the complexity of each tree. These parameters were included to compare shallow and deeper ensemble structures.

TABLE IV
HYPERPARAMETER GRID USED FOR MULTILAYER PERCEPTRON.

Hyperparameter	Values
Hidden layer sizes	$\{(32), (64), (32, 16)\}$
Activation function	relu
Regularization parameter α	$\{0.0001, 0.001, 0.01\}$
Learning rate	adaptive
Maximum iterations	1000

For MLP, different hidden layer sizes were tested to compare smaller and larger neural architectures. The regularization parameter α was varied to control overfitting. The ReLU activation function and adaptive learning rate were used across all MLP configurations.

B. Calibration Hyperparameters

Post-hoc calibration was applied after model training using the calibration split. The base classifier was not retrained during calibration. The calibration methods and their tested hyperparameters are shown in Table V.

TABLE V
HYPERPARAMETERS USED FOR POST-HOC CALIBRATION METHODS.

Calibration Method	Hyperparameters
Platt Scaling	sigmoid calibration
Isotonic Calibration	isotonic regression
Histogram Binning	bins $= \{5, 10, 15\}$
Histogram Binning Strategy	{uniform, quantile}

Platt Scaling was used as a parametric calibration method, while isotonic calibration was used as a more flexible non-parametric method. For histogram binning, different numbers of bins and binning strategies were tested to analyze how the granularity of probability intervals affects calibration performance.

C. Bootstrap Hyperparameters

The uncertainty decomposition experiment used bootstrap ensembles. For each selected model architecture and feature subset, the best hyperparameter configuration was first selected using the validation composite score. Then, bootstrap models were trained using the settings shown in Table VI.

TABLE VI
HYPERPARAMETERS USED FOR BOOTSTRAP-BASED UNCERTAINTY DECOMPOSITION.

Parameter	Value
Number of bootstrap models M	30
Sampling method	With replacement
Training source	Original training split
Evaluation split	Test split
Selection criterion	Validation composite score

The bootstrap ensemble was used to estimate total, aleatoric, and epistemic uncertainty. Aleatoric uncertainty was computed from the average entropy of individual bootstrap predictions, while epistemic uncertainty was computed from the difference between total uncertainty and aleatoric uncertainty.